Because we evaluate our models on papers published in 2022, the likelihood of test papers appearing in the pretraining corpora for the models is substantially reduced. We additionally performed a manual examination of GPT-4 memorization in our gold set based on 2022 ACL Anthology papers, by seeing if GPT-4 could complete information such as method names or generate text that strongly mimics the ground truth papers, and found no evidence of this occurring. The Meditron-7b (Chen et al., 2023) uses PubMed with a cut-off in August 2023, and our biochemical test set only includes PubMed papers after 2023/08.

## Acknowledgements

This work is supported by the Molecule Maker Lab Institute: an AI research institute program supported by NSF under award No. 2019897, by DOE Center for Advanced Bioenergy and Bioproducts Innovation U.S. Department of Energy, Office of Science, Office of Biological and Environmental Research under Award Number DESC0018420, by U.S. the AI Research Institutes program by National Science Foundation and the Institute of Education Sciences, Department of Education through Award No. 2229873 - AI Institute for Transforming Education for Children with Speech and Language Processing Challenges, and by AI Agriculture: the Agriculture and Food Research Initiative (AFRI) grant no. 2020-67021- 32799/project accession no.1024178 from the USDA National Institute of Food and Agriculture. The views and conclusions contained herein are those of the authors and should not be interpreted as necessarily representing the official policies, either expressed or implied of, the National Science Foundation, the U.S. Department of Energy, and the U.S. Government. The U.S. Government is authorized to reproduce and distribute reprints for governmental purposes notwithstanding any copyright annotation therein.

## References

Daniil A Boiko, Robert MacKnight, Ben Kline, and Gabe Gomes. 2023. Autonomous chemical research with large language models. *Nature*, 624(7992):570–578.

Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel Ziegler, Jeffrey Wu, Clemens Winter, Chris Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. 2020. Language models are few-shot learners. In *Advances in Neural Information Processing Systems*, volume 33, pages 1877–1901. Curran Associates, Inc.

Nicholas Carlini. 2023. A llm assisted exploitation of ai-guardian. *Cryptography and Security Repository*, arXiv:2307.15008.

Nicholas Carlini, Daphne Ippolito, Matthew Jagielski, Katherine Lee, Florian Tramer, and Chiyuan Zhang. 2023. Quantifying memorization across neural language models. In *The Eleventh International Conference on Learning Representations*.

Arie Cattan, Sophie Johnson, Daniel S Weld, Ido Dagan, Iz Beltagy, Doug Downey, and Tom Hope. 2021. Scico: Hierarchical cross-document coreference for scientific concepts. In *3rd Conference on Automated Knowledge Base Construction*.

Zeming Chen, Alejandro Hernández Cano, Angelika Romanou, Antoine Bonnet, Kyle Matoba, Francesco Salvi, Matteo Pagliardini, Simin Fan, Andreas Köpf, Amirkeivan Mohtashami, et al. 2023. Meditron-70b: Scaling medical pretraining for large language models. *Computation and Language Repository*, arXiv:2311.16079.

Arman Cohan, Iz Beltagy, Daniel King, Bhavana Dalvi, and Dan Weld. 2019. Pretrained language models for sequential sentence classification. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 3693–3699, Hong Kong, China. Association for Computational Linguistics.

Cyprien de Masson d'Autume, Sebastian Ruder, Lingpeng Kong, and Dani Yogatama. 2019. Episodic memory in lifelong language learning. In *Advances in Neural Information Processing Systems*, volume 32. Curran Associates, Inc.

Qian Dong, Yaoming Zhu, Mingxuan Wang, and Lei Li. 2022. Learning when to translate for streaming speech. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 680–694.

Sam Henry and Bridget T McInnes. 2017. Literature based discovery: models, methods, and trends. *Journal of biomedical informatics*, 74:20–32.

Tom Hope, Doug Downey, Oren Etzioni, Daniel S Weld, and Eric Horvitz. 2023. A computational inflection for scientific discovery. *Communications of the ACM*.

Zhe Hu, Zuohui Fu, Yu Yin, and Gerard de Melo. 2021. Context-aware interaction network for question matching. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 3846–3853, Online and Punta Cana, Dominican Republic. Association for Computational Linguistics.

Qian Huang, Jian Vora, Percy Liang, and Jure Leskovec. 2023. Benchmarking large language models as ai research agents. *Machine Learning Repository*, arXiv:2310.03302.

Ting-Hao Kenneth Huang, Chieh-Yang Huang, Chien-Kuang Cornelia Ding, Yen-Chia Hsu, and C. Lee Giles. 2020. CODA-19: Using a non-expert crowd to annotate research aspects on 10,000+ abstracts in the COVID-19 open research dataset. In *Proceedings of the 1st Workshop on NLP for COVID-19 at ACL 2020*, Online. Association for Computational Linguistics.

Rezarta Islamaj, Robert Leaman, Sun Kim, Dongseop Kwon, Chih-Hsuan Wei, Donald C. Comeau, Yifan Peng, David Cissel, Cathleen Coss, Carol Fisher, Rob Guzman, Preeti Gokal Kochar, Stella Koppel, Dorothy Trinh, Keiko Sekiya, Janice Ward, Deborah Whitman, Susan Schmidt, and Zhiyong Lu. 2021. Nlm-chem, a new resource for chemical entity recognition in pubmed full text literature. *Scientific Data*, 8(1):91.

Bernal Jimenez Gutierrez, Nikolas McNeal, Clayton Washington, You Chen, Lang Li, Huan Sun, and Yu Su. 2022. Thinking about GPT-3 in-context learning for biomedical IE? think again. In *Findings of the Association for Computational Linguistics: EMNLP 2022*, pages 4497–4512, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.

Po-Ting Lai, Chih-Hsuan Wei, Ling Luo, Qingyu Chen, and Zhiyong Lu. 2023. Biorex: Improving biomedical relation extraction by leveraging heterogeneous datasets. *Journal of Biomedical Informatics*, 146:104487.

Chin-Yew Lin. 2004. ROUGE: A package for automatic evaluation of summaries. In *Text Summarization Branches Out*, pages 74–81, Barcelona, Spain. Association for Computational Linguistics.

Jiachang Liu, Dinghan Shen, Yizhe Zhang, Bill Dolan, Lawrence Carin, and Weizhu Chen. 2022. What makes good in-context examples for GPT-3? In *Proceedings of Deep Learning Inside Out (DeeLIO 2022): The 3rd Workshop on Knowledge Extraction and Integration for Deep Learning Architectures*, pages 100–114, Dublin, Ireland and Online. Association for Computational Linguistics.

Kyle Lo, Lucy Lu Wang, Mark Neumann, Rodney Kinney, and Daniel Weld. 2020. S2ORC: The semantic scholar open research corpus. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4969–4983, Online. Association for Computational Linguistics.

Ilya Loshchilov and Frank Hutter. 2019. Decoupled weight decay regularization. In *Proceedings of the 7th International Conference on Learning Representations*.

Yi Luan, Luheng He, Mari Ostendorf, and Hannaneh Hajishirzi. 2018. Multi-task identification of entities, relations, and coreference for scientific knowledge graph construction. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 3219–3232, Brussels, Belgium. Association for Computational Linguistics.

Ling Luo, Chih-Hsuan Wei, Po-Ting Lai, Robert Leaman, Qingyu Chen, and Zhiyong Lu. 2023. AIONER: all-in-one scheme-based biomedical named entity recognition using deep learning. *Bioinformatics*, 39(5):btad310.

Rahul Nadkarni, David Wadden, Iz Beltagy, Noah A Smith, Hannaneh Hajishirzi, and Tom Hope. 2021. Scientific language models for biomedical knowledge base completion: an empirical study. *AKBC*.

Mark Neumann, Daniel King, Iz Beltagy, and Waleed Ammar. 2019. ScispaCy: Fast and robust models for biomedical natural language processing. In *Proceedings of the 18th BioNLP Workshop and Shared Task*, pages 319–327, Florence, Italy. Association for Computational Linguistics.

Allen Newell and Herbert Simon. 1956. The logic theory machine–a complex information processing system. *IRE Transactions on information theory*, 2(3):61–79.

Harsha Nori, Nicholas King, Scott Mayer McKinney, Dean Carignan, and Eric Horvitz. 2023. Capabilities of GPT-4 on medical challenge problems. *Computation and Language Repository*, arXiv:2303.13375.

Aaron van den Oord, Yazhe Li, and Oriol Vinyals. 2018. Representation learning with contrastive predictive coding. *Machine Learning Repository*, arXiv:1807.03748.

OpenAI. 2023. Gpt-4 technical report. *Computation and Language Repository*, arXiv:2303.08774.

Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. 2022. Training language models to follow instructions with human feedback. *Advances in Neural Information Processing Systems*, 35:27730–27744.

Yujia Qin, Jiajie Zhang, Yankai Lin, Zhiyuan Liu, Peng Li, Maosong Sun, and Jie Zhou. 2022. ELLE: Efficient lifelong pre-training for emerging data. In *Findings of the Association for Computational Linguistics: ACL 2022*, pages 2789–2810, Dublin, Ireland. Association for Computational Linguistics.

Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. 2020. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21(140):1–67.

Nils Reimers and Iryna Gurevych. 2019. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 3982–3992, Hong Kong, China. Association for Computational Linguistics.

Herbert A Simon. 1973. Does scientific discovery have a logic? *Philosophy of science*, 40(4):471–480.

Daniel N Sosa and Russ B Altman. 2022. Contexts and contradictions: a roadmap for computational drug repurposing with knowledge inference. *Briefings in Bioinformatics*, 23(4):bbac268.

Don R Swanson. 1986. Undiscovered public knowledge. *The Library Quarterly*, 56(2):103–118.

Justin Sybrandt, Ilya Tyagin, Michael Shtutman, and Ilya Safro. 2020. Agatha: automatic graph mining and transformer based hypothesis generation approach. In *Proceedings of the 29th ACM International Conference on Information & Knowledge Management*, pages 2757–2764.

Vahe Tshitoyan, John Dagdelen, Leigh Weston, Alexander Dunn, Ziqin Rong, Olga Kononova, Kristin A Persson, Gerbrand Ceder, and Anubhav Jain. 2019. Unsupervised word embeddings capture latent knowledge from materials science literature. *Nature*, 571(7763):95–98.

Qingyun Wang, Lifu Huang, Zhiying Jiang, Kevin Knight, Heng Ji, Mohit Bansal, and Yi Luan. 2019. PaperRobot: Incremental draft generation of scientific ideas. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 1980–1991, Florence, Italy. Association for Computational Linguistics.

Alex Warstadt, Aaron Mueller, Leshem Choshen, Ethan Wilcox, Chengxu Zhuang, Juan Ciro, Rafael Mosquera, Bhargavi Paranjabe, Adina Williams, Tal Linzen, et al. 2023. Findings of the babylm challenge: Sample-efficient pretraining on developmentally plausible corpora. In *Proceedings of the BabyLM Challenge at the 27th Conference on Computational Natural Language Learning*, pages 1–34.

Chih-Hsuan Wei, Alexis Allot, Kevin Riehle, Aleksandar Milosavljevic, and Zhiyong Lu. 2022. tmvar 3.0: an improved variant concept recognition and normalization tool. *Bioinformatics*, 38(18):4449–4451.

Chih-Hsuan Wei, Ling Luo, Rezarta Islamaj, Po-Ting Lai, and Zhiyong Lu. 2023. GNorm2: an improved gene name recognition and normalization system. *Bioinformatics*, 39(10):btad599.

Thomas Wolf, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Moi, Pierric Cistac, Tim Rault, Remi Louf, Morgan Funtowicz, Joe Davison, Sam Shleifer, Patrick von Platen, Clara Ma, Yacine Jernite, Julien Plu, Canwen Xu, Teven Le Scao, Sylvain Gugger, Mariama Drame, Quentin Lhoest, and Alexander Rush. 2020. Transformers: State-of-the-art natural language processing. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 38–45, Online. Association for Computational Linguistics.

Yi Xu, Shuqian Sheng, Bo Xue, Luoyi Fu, Xinbing Wang, and Chenghu Zhou. 2023. Exploring and verbalizing academic ideas by concept co-occurrence. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 13001–13027, Toronto, Canada. Association for Computational Linguistics.

Deming Ye, Yankai Lin, Peng Li, and Maosong Sun. 2022. Packed levitated marker for entity and relation extraction. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 4904–4917, Dublin, Ireland. Association for Computational Linguistics.

Weizhe Yuan, Graham Neubig, and Pengfei Liu. 2021. BARTScore: Evaluating generated text as text generation. In *Advances in Neural Information Processing Systems*.

Tianyi Zhang*, Varsha Kishore*, Felix Wu*, Kilian Q. Weinberger, and Yoav Artzi. 2020. Bertscore: Evaluating text generation with bert. In *Proceedings of the 8th International Conference on Learning Representations*.

Ran Zhou, Xin Li, Ruidan He, Lidong Bing, Erik Cambria, Luo Si, and Chunyan Miao. 2022. MELM: Data augmentation with masked entity language modeling for low-resource NER. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 2251–2262, Dublin, Ireland. Association for Computational Linguistics.

# A Dataset Collection

## A.1 NLP Dataset Collection

We download ACL Anthology papers from 1952 to 2022 using Semantic Scholar Academic Graph API.[11] We filter out papers without abstracts and not written in English to obtain 67,408 papers. Our dataset has 58,874 papers before 2021, 5,946 papers from 2021, and 2,588 from 2022. We first use PL-Marker (Ye et al., 2022) pretrained on SciERC (Luan et al., 2018) to extract nodes belonging to six types: *Task*, *Method*, *Evaluation Metric*, *Material*, *Other Scientific Terms*, and *Generic*

[11] www.semanticscholar.org/product/api